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Impact of Diwali celebrations on urban air and noise quality in Delhi City, India

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Abstract A study was conducted in the residential areas of Delhi, India, to assess the variation in ambient air quality and ambient noise levels during pre-Diwali month (DM), Diwali day (DD) and post-Diwali month during the period 2006 to 2008. The use of fireworks during DD showed 1.3 to 4.0 times increase in concentration of respirable particulate matter (PM_{10}) and 1.6 to 2.5 times increase in concentration of total suspended particulate matter (TSP) than the concentration during DM. There was a significant increase in sulfur dioxide (SO_2) concentration but the concentration of nitrogen dioxide (NO_2) did not show any considerable variation. Ambient noise level were 1.2 to 1.3 times higher than normal day. The study also showed a strong correlation between PM_{10} and TSP ($R^2 \geq 0.9$) and SO_2 and NO_2 ($R^2 \geq$ 0.9) on DD. The correlation between noise level and gaseous pollutant were moderate ($R^2 \geq 0.5$). The average concentration of the pollutants during DD was found higher in 2007 which could be due to adverse meteorological conditions. The statistical interpretation of data indicated that the celebration of Diwali festival affects the ambient air and noise quality. The study would provide public awareness about the health risks associated with the celebrations of Diwali festival so as to take proper precautions.

Keywords Air pollutants · Ambient noise · Diwali · Fireworks · Correlation · Regression

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Introduction

Urban atmospheric air and noise pollution are receiving increased attention in present years due to their significant impact on human health and environment. People living in urban areas are exposed to complex mixture of environmental pollutants due to heterogeneous and spatial distribution of emission sources and meteorological conditions. According to the World Health Organization, noise pollution is the third most hazardous environmental pollution (WHO 2005). The potential sources of both the pollutants are closely related to urbanization, industrialization, growth of population and its associated demand on transportation. The urban air and noise pollution in Delhi have increased significantly and exceeded the level of tolerance. Presently more than 1,300 tonnes of pollutants are emitted by the vehicles plying in Delhi (Goyal and Sidhartha

2003). The contribution of air pollution from different sources in Delhi is indicated in Table 1. It has been estimated that hundreds of thousands of cases of respiratory illness are associated with atmospheric pollution each year in Delhi (Faiz and Sturm 2000). To protect urban air quality in Delhi, it is necessary to reduce the steep increase of vehicular population, associated its vehicular emission and traffic annoyances. The social survey data has shown that noise annoyance and sleep disturbances are considered to be the most important noise effect. Noise annoyance, a feeling of displeasure-irritation or disturbance, gives a negative effect on the community or individual (Jakovljevic et al. 2009). Indeed studies have shown that noise annoyance from transportation produces a variety of negative emotions including anger, disappointment, unhappiness, anxiety and even depression (Murphy et al. 2009).

Diwali, the festival of lights in India, is regarded as one of the most important and glamorous festivals among all the festivals celebrated in India. It is generally celebrated during post-monsoon season (September–November) every year. Meteorological parameters e.g., temperature, relative humidity, wind speed and mixing height recorded during Diwali festival day (2006–2008) are presented in Table 2. The lighting of fire crackers with brighter sparkles, louder noise, aesthetic forms of light, seems most essential and appropriate in a Diwali festival. The temporary joy and happiness leads to intense hazardous air and noise pollution. The crackers generally consist of several chemicals, which are harmful to the human health. These chemicals are copper, cadmium, lead, magnesium, sodium, zinc, nitrate and nitrite. The toxic substances used to prepare the crackers release toxic gases and fine particulate matters which are closely associated with severe health diseases especially to the infants, women and elderly people. The present study was aimed to evaluate the following:

- Short-term variation of ambient air quality and ambient noise level during DD
- Comparison of ambient air quality during DD and DM
- Comparison of ambient noise during DD and normal day (ND)

Table 1 Contribution of air pollution from different sources in Delhi

Source	1970–1971 (%)	1980–1981 (%)	1990–1991 (%)	2000–2001 (%)
Industrial	56	40	29	70
Vehicular	23	42	64	72
Domestic	21	18	7	8

CPCB (Guidelines for AAQM 2005)

Table 2 Twenty-four hourly average of meteorological parameters recorded during DD

Year	Episode	Temperature (°C)	RH (%)	Wind velocity (m/s)	Mixing height (m)
2006	21 October 2006	24.5	70.13	0.29	467
2007	09 November 2007	21.1	86.63	0.1	301
2008	28 October 2008	23.8	63.50	0.29	446

http://cpcb.nic.in/

Present scenario of ambient air pollution and its health problem to the resident of Delhi

Delhi (28°39′ N, 77°13′ E) is located in north of India and approximately 216 m above mean sea level. The present population of the city in the year 2010 is near about 18 million and covered area is approximately 1,500 km^2. The climate of the city is mainly influenced by its inland position, ranging between arid and semi-arid with maximum temperature of 46°C in summer and minimum 1°C in winter. The average annual rainfall intensity is approximately 670 mm. The micrometeorological data regarding wind speed and wind direction (WD) during the study period was collected from Meteorological Department of Delhi. The wind roses of post-monsoon season of Delhi are shown in Fig. 1. During the post-monsoon season, the predominant WD was from north-west direction in which the calm conditions varied from 48.63% to 62.50%. Existing industries, power plant, vehicular activities and frequent dust storm mainly contributed high concentration of air pollutants in Delhi (Chelani and Devotta 2007). The vehicular movements on road

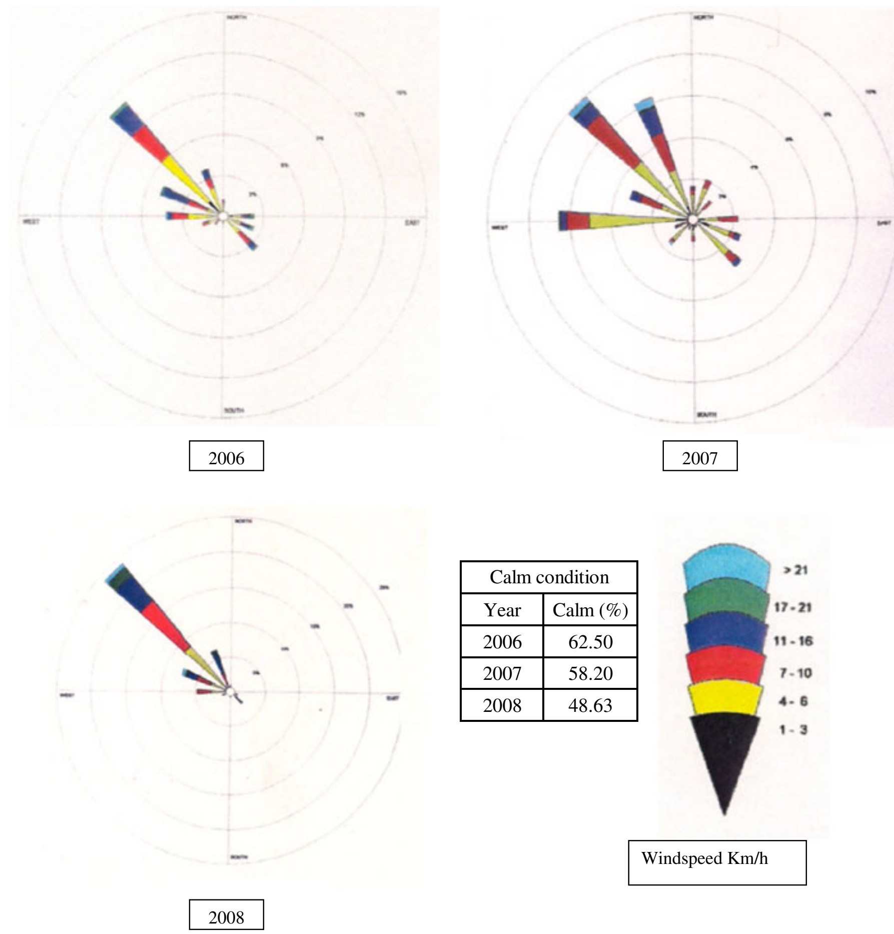

Fig. 1 Wind roses of post-monsoon season for Delhi during 2006–2008. Source: Meteorological Department of Delhi, 2006–2008

get restricted due to shortage of road space and unplanned traffic management, increasing exhaust emission and noise annoyance.

It was estimated that out of every ten school children in Delhi, one suffers from asthma (Cropper et al. 1997). Central Pollution Control Board (CPCB), New Delhi, in collaboration with All India Institute of Medical Sciences, New Delhi, conducted a health survey during 1997–1998 on individuals residing in the residential areas of Delhi. Several medical symptoms were prevalent among Delhi residents and its percentage due to multiple environmental pollutions is provided in Table 3. Eye irritation was the most common complaint which was 44% of the total surveyed people. The next was cough and pharyngitis, which were approximately 28.8% and 16.5%, respectively. Sixteen percent of Delhi residents were affected by dyspnea. Symptoms like nausea (10%) and respiratory problems (5.9%) were some of the other complaints. Respiratory problems (bronchitis, dyspnea, and rhinitis) and nervous system problems (headaches and depression) were common and widespread. Even the lung functions, such as forced expiratory volume (FEV1), forced vital capacity (FVC) and peak expiratory flow (PEF) were found to be much lower than the Clement Clarks standard probably due to high exposure of air pollution and traffic annoyance as presented in Table 4. It was also observed that non-smoking people in Delhi have more risk to acute and chronic illness among all age groups by urban air pollutants than smokers in Delhi due to urban pollution.

Table 3 Medical symptoms prevalent among Delhi residents

Name of disease	Male $N = 788$	Female $N = 533$	Total $N = 1{,}321$	Percent
Irritation of eye	354	233	587	44.4
Cough	224	157	381	28.8
Pharyngitis	138	81	219	16.5
Dyspnea	117	97	214	16.2
Headache	78	114	192	14.5
Nausea	50	82	132	10
Vomiting	44	79	123	9.3
Conjunctivitis	59	47	106	8
Abdominal pain	36	50	86	6.5
Respiratory problems	51	27	78	5.9
Rhinitis	23	21	44	3.3
Bronchitis	17	13	30	2.3
Burning of mouth and throat	8	3	11	0.8
Epistaxis	2	4	6	0.5
Depression	2	–	2	0.2
Non-smokers	–	–	–	87
Smokers	–	–	–	13

CPCB (Parivesh September, 2001)

Table 4 Comparison of lung function (spirometry) of Delhi residents with Clement Clark's Standard

Age in years	FEV1 (L/s)		FVC (L)		FEV1/FVC (ratio)		PEF (L/min)	
	1	2	1	2	1	2	1	2
20–30	4.67	2.1	5.5	2.3	82.7	92.6	619	236.6
31–40	4.38	1.98	5.3	2.2	80.4	92	593	223.9
41–50	4.09	1.95	5.2	2.2	78.5	90.6	566	235.4
51–60	3.8	1.58	4.8	1.9	76.8	80.1	541	182.3
> 60	3.5	1.16	4.5	1.5	75	83.2	516	146.2

CPCB (Parivesh September, 2001)
1 standard, 2 observed, *FEV1* forced expiratory volume (these measures the amount of air forcefully exhaled in a sustained breath), *FVC* forced vital capacity (this is the measure of the maximum amount of air exhaled after inhaling as deeply as one can), *PEF* peak expiratory flow (this measures how fast one can exhale the air after its inhalation)

Table 5 Average concentration of air pollutants in pre-DM and post-DM at Pitampura residential area

Year	Month	SO_2	NO_2	PM_{10}	TSP
2006	September 2006 (pre-DM)	8 ± 6	25 ± 8	97 ± 31	328 ± 82
	November 2006 (post-DM)	5 ± 1	37 ± 10	176 ± 90	439 ± 200
2007	October 2006 (pre-DM)	4 ± 0	22 ± 5	203 ± 23	369 ± 76
	December 2006 (post-DM)	4 ± 0	37 ± 2	235 ± 64	483 ± 112
2008	September 2006 (pre-DM)	4 ± 0	36 ± 5	101 ± 34	254 ± 85
	November 2006 (post-DM)	6 ± 2	55 ± 24	227 ± 48	519 ± 135

http://cpcb.nic.in/
$\pm$ standard deviation

Table 6 National ambient air quality standards (CPCB 2000a, b)

Pollutants	Concentration of ambient air ($\mu g/m^3$)		
	Industrial area	Residential rural and other areas	Sensitive area
SO_2	120	80	30
NO_2	120	80	30
PM_{10}	150	100	75
TSP	500	200	100

Time weighted average 24 h. Source: http://cpcb.nic.in/

Pitampura is one of the prime residential areas of North Delhi, India. The area is encompassed between outer and inner ring roads, National Highway of India and Rohtak road. CPCB has a monitoring station at Pitampura and the data generated has been used in this study.

Methodology

Monitoring was carried out using high volume samplers and respirable dust samplers. Particulate matter e.g., TSP and PM_{10} were estimated by gravimetric method. A known amount of air was drawn through pre-weighed glass fiber filter paper (GF/A) at a flow rate of 0.8–1.3 m^3/min. Gaseous pollutants namely SO_2 and NO_2 were measured by Improved West and Geake and Jacob and Hochheiser methods respectively. Concentration of the pollutants were measured in microgram per cubic meter. Noise level were measured using real-time analyzer method on hourly basis.

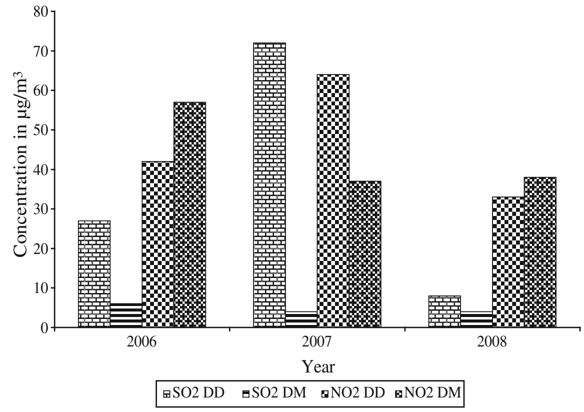

Fig. 3 Variation of concentrations of SO_2 and NO_2 during DD and DM during 2006 to 2008. Source: http://cpcb.nic.in/

Results and discussion

The average ambient concentration of TSP, PM_{10}, SO_2 and NO_2 in pre- and post-DM at Pitampura residential area is presented in Table 5 indicating that the concentration of SO_2 and NO_2 during pre- and post-DM were within the permissible limit of National Ambient Air Quality Standards (NAAQS) of CPCB 2000 presented in

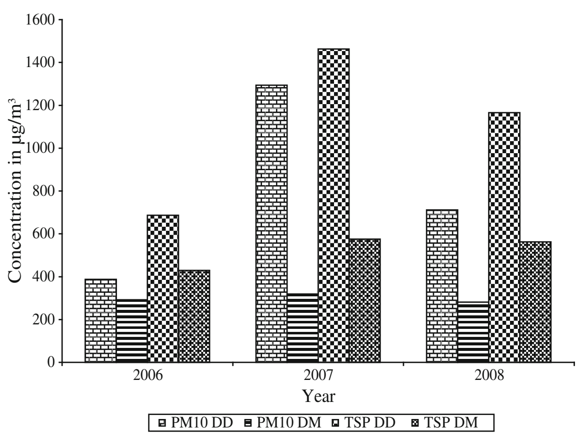

Fig. 2 Variation of concentration of TSP and PM_{10} during DD and DM during 2006 to 2008. Source: http://cpcb.nic.in/

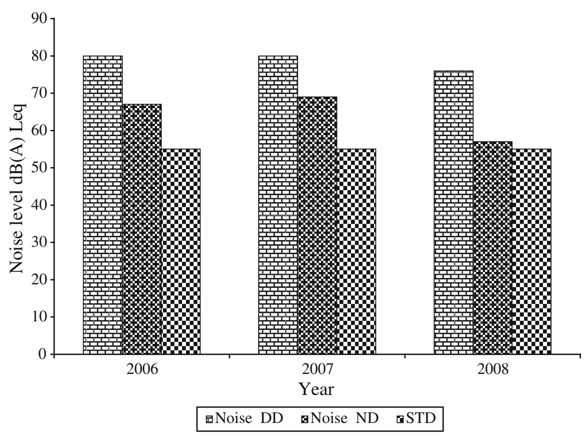

Fig. 4 Variation of noise level during DD and ND during 2006 to 2008. Source: http://cpcb.nic.in/

Table 7 Ambient air quality standards in respect of noise

Area code	Category of area/zone	Day time dB(A) Leq	Night time dB(A) Leq
A	Industrial area	75	70
B	Residential area	55	45
C	Commercial area	65	55
D	Silence zone	50	40

http://cpcb.nic.in/

Day time 6:00 a.m. to 10:00 p.m., *night time* 10:00 p.m. to 6:00 a.m., *dB (A) Leq* the time weighted average of the level of sound in decibels on scale A, which is related to human hearing

Table 6. The TSP and PM_{10} concentration were alarmingly high as compared to NAAQS which could be attributed due to large-scale construction activity, traffic movement, soil dust and paved road dust. The average ambient concentration of TSP and PM_{10} and SO_2 and NO_2 during DD and DM are presented in Figs. 2 and 3. The TSP concentration during DD was 1.6 to 2.5 times higher than DM. The PM_{10} concentration during DD was 1.3 to 4 times higher than DM during the study period. Fireworks during Diwali festival led to a short-term variation of air quality and two to three times increase in PM_{10} and TSP concentration in Hissar City, India (Ravindra et al. 2003). The average ambient concentration of NO_2 during DD was lesser than DM in the year 2006 and 2008. In the year 2007, the average NO_2 concentration during DD was 1.7 times higher than DM which might be due to adverse metrological conditions in Delhi. Although SO_2 concentration during DD was within the permissible limit of NAAQS, but it was 2 to 18 times higher as compared to DM in the residential areas, during 2006 to 2008. High levels of SO_2 are particularly dangerous in the presence of PM because it slowly adsorbs on fine atmospheric particles and can be transported very deep into the lungs and therefore can stay there for a long time (Singh et al. 2009). However, PM especially PM_{10}, SO_2 and NO_2 are of special interest, as they are known to be potentially injurious to the respiratory passages and can cause other health effects (Bull et al. 2001). The health problems become severe due to combined effect of urban air pollution and ambient noise.

The average ambient noise levels during DD and ND at Pitampura residential area are shown in Fig. 4. The average ambient noise level during ND were ranging from 57 to 69 dB (A) Leq but noise level varied from 76 to 80 dB (A) Leq on DD. The ambient air quality standards in respect of noise are presented in Table 7. The ambient noise level was above the permissible limit of CPCB 2000 even at ND also. The Pearson correlation coefficient and linear regression analysis were also performed between measured pollutants to understand the relationship among air pollutants and relationship between air pollutants and ambient noise in detail. The summary of ambient air and noise pollutants during DD and DM is presented in Table 8. It was noticed that linear regression of SO_2/NO_2 and PM_{10}/TSP has strong relationship ($R^2 \geq 0.9$) during DD. The

Table 8 Summary of ambient air and noise pollutants during DD and DM

Parameter	DD			DM		
	Correlation	Regression	R^2	Correlation	Regression	R^2
(SO_2/NO_2)	0.99	$NO_2 = 0.49$ SO_2 +29	0.99	0.99	$NO_2 = 9.75$ SO_2 −1.50	0.99
(PM_{10}/TSP)	0.96	$PM_{10} = 1.12$ TSP −441	0.91	0.31	$PM_{10} = 0.08$ TSP +257.72	0.10
(Leq/SO_2)	0.73	$SO_2 = 10.38$ Leq −781	0.53	0.36	$SO_2 = 0.06$ Leq +0.52	0.13
(Leq/NO_2)	0.72	$NO_2 = 5$ Leq −347	0.52	0.32	$NO_2 = 0.56$ Leq +8.20	0.10

http://cpcb.nic.in/

R^2 regression coefficient

linear regression of Leq between SO_2 and NO_2 showed moderate relationship ($R^2 \geq 0.5$) during DD. However, during DM, only SO_2/NO_2 showed strong relationship ($R^2 \geq 0.9$).

Conclusions

The study illustrated that the celebrations of Diwali festival are contributing higher concentration of air pollutants, which are one of the additional major sources of air pollution during DM other than the existing sources. The mixing height is one of the important meteorological parameters as low mixing height accumulates the pollutants at lower level thereby increasing the concentration of air pollutants. The concentration of ambient air quality as well as noise level on DD during 2006 and 2008 were found to be lower as compared to DD during 2007 which could be due to use of lesser number of fireworks and to some extent the favorable meteorological conditions, such as increased mixing height and temperature. The present study would be helpful to assess the impact of Diwali celebrations on urban air and noise quality and to provide public awareness as well as suggest mitigation measures to curb the health risks associated with the celebrations of Diwali festival.

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